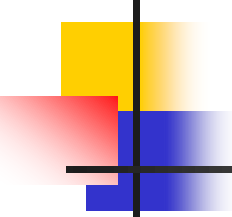




ET 8311: Digital Telecommunications

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Mode of Assessment

- Continuous Assessment (CA) + University Examination (UE)

CA – 40% of Final

- | | |
|--------------|-----|
| ❑ Test 1 | 15% |
| ❑ Test 2 | 15% |
| ❑ Assignment | 10% |

UE – 60% of Final

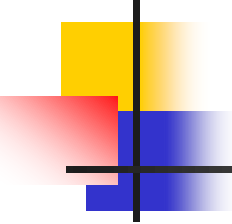
Final = CA + UE

- Number of credit.....9



Outcome

- Apply fundamental concepts of digital telecommunication systems.
- Analyze baseband transmission of digital signals.
- Characterize time-domain signals using geometric representations.
- Analyze and design passband digital communication systems.
- Analyze and design digital modulation schemes to solve complex problems of telecommunications.

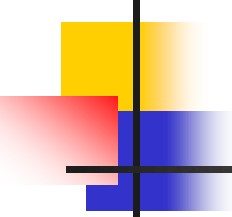


Course Outline

- Lecture 1: Introduction

- Overview of Digital Telecommunication Systems

- Basic elements of digital communication systems
 - Merits and demerits of digital communication systems



Cont.

❑ Lecture 2: Pulse Modulation

- ❑ Evolution of public switched telecommunication
- ❑ Networks stronger exchange, Crossbar exchange, Stored program control exchange, Digital exchange
- ❑ Basic telecommunication equipment, Telephone handset, hybrid circuit, echo suppressors and cancellers, PCM coders, modems and relays.



Cont.

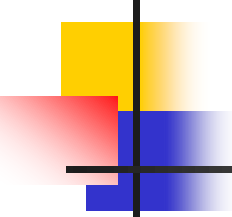
□ Lecture 3: Signal Space Analysis

- Introduction
- Geometric representation of signals
- Coherent detection
- correlation receiver
- matched filter and probability error



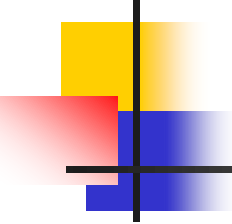
Cont.

- ❑ Lecture 4: Baseband digital transmission
 - ❑ Intersymbol interference (ISI),
 - ❑ Nyquist's Criterion for distortion-less baseband binary transmission
 - ❑ Correlative-level coding
 - ❑ Baseband M-ary PAM transmission,
 - ❑ Optimum linear receiver and equalization.



Cont.

- ❑ Lecture 5: Passband digital transmission
 - ❑ Passband transmission model
 - ❑ coherent phase shift keying
 - ❑ Hybrid amplitude/phase modulation schemes
 - ❑ Coherent frequency shift keying
 - ❑ Differential phase shift keying
 - ❑ Other modulation techniques (e.g. CPFSK, MSK, GMSK etc)
 - ❑ Comparison of digital modulation schemes using single carrier.



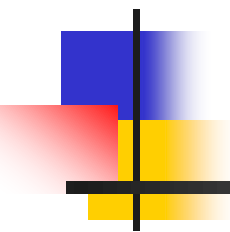
Cont.

- ❑ Lecture 6: Synchronization
 - ❑ Synchronization levels
 - ❑ Scrambling
 - ❑ Phase-locked loop (PLL)
 - ❑ Carrier recovery
 - ❑ Symbol synchronization



References

- Ali Grami, (2015) "Introduction to Digital Communications", Elsevier Science, ISBN0124076580, 9780124076587
- Simon Haykin (2001) "Communications systems" John Willey & Sons, Inc, 4th Edition ISBN0-471-17868-1
- B. Sklar, (2001) "Digital Communications: Fundamentals and Applications", Prentice Hall, 2nd ed.,
- J. G. Proakis, (2000) "Digital Communications", McGraw Hill, 4th ed.,
- B.P. Lathi, (2006) "Modern Digital and Analog Communication Systems", Oxford University Press, 3rd Ed.
- Leon W. Couch, (2006) "Digital and Analog Communication Systems", Discovery Press, 7th ed..
- M. S. Roden, (2003) "Analog and Digital Communication Systems", Discovery Press, 5th Ed.



Thank You!